

Akarshan Ghosh

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Summary

IoT and Embedded Systems developer with hands-on experience in computer vision, edge AI, and full-stack IoT platforms. Worked with IIT Guwahati TIDF and Indian Railways to build operational prototypes involving multi-sensor telemetry, cloud architectures, and real-time monitoring.

Technical Skill

- **Languages:** Python, Embedded C, JavaScript, TypeScript
- **Machine Learning / CV:** CNNs, OpenCV, Multi-Object Tracking, Behavior Analysis
- **Backend:** FastAPI, Node.js, Express.js, Django
- **Frontend:** React.js, Tailwind CSS
- **Databases:** MongoDB, MySQL, SQLite
- **Cloud / DevOps:** AWS, Cloud Deployments, ThingSpeak
- **Tools:** Git, Thunder Client, Postman, Figma

Experience

Intern, IIT Guwahati – TIDF

October 2024 – November 2025

- Developed a cloud-based MERN application used by Railway staff and IIT TIDF engineers to monitor train emergency brakes and telemetry during prototype deployment.
- Designed a scalable per-coach ingestion pipeline processing 9 telemetry fields (GPS, chain status, temperature, humidity, memory, etc.) at a 5-second refresh rate.
- Implemented secure authentication and real-time dashboards for 5 coaches, reducing manual logging and improving operational monitoring efficiency.
- Optimized backend APIs using Node.js and Express.js for reliable cloud ingestion and stable performance.
- Tech Used: React, Node.js, Express.js, MongoDB Atlas, Tailwind CSS, Thunder Client.

Intern, IIT Guwahati – TIDF

June 2024 – July 2024

- Built an IoT flood monitoring prototype using NodeMCU ESP8266, sonar sensor, and GSM with a 5-second update interval for real-time hazard alerts.
- Integrated cloud dashboards via Django, SQL, and ThingSpeak for live visualization and field-tested data monitoring.

Projects

EYEORA – Intelligent AI-Based Surveillance System

[GitHub Link](#)

- Engineered a real-time surveillance platform with multi-object detection and tracking, behavior analysis (idle, window shopping, buying, threat), and weapon detection (knife/blade) running at 20–25 FPS on GTX 1650 (4GB).
- Supported multi-stream ingestion for simultaneous real-time inference.
- Built a cloud-ready FastAPI backend for modular, scalable analytics and automated reporting.
- Tech Used: Python, CNNs, OpenCV, FastAPI, Deep Learning, Computer Vision, IP Cameras.

E-Chain Tracker MERN App (Indian Railways)

[GitHub Link](#)

- Developed a cloud-deployed MERN application enabling Railway staff to monitor 5 coaches with 5-second IoT refresh intervals.
- Visualized 9 telemetry fields including GPS coordinates, chain status, temperature, humidity, memory usage, and error flags.

- Implemented secure authentication, responsive coach dashboards, and train search functionality to improve operational monitoring efficiency.
- Tech Used: JavaScript, React, Node.js, Express.js, MongoDB, Tailwind CSS, Thunder Client.

Flood Monitoring System Web Application

[GitHub Link](#)

- Built a Django-based flood monitoring dashboard with 5-second ThingSpeak polling for real-time water level updates.
- Implemented interactive frontend using JavaScript and Tailwind CSS to support internship field tests.
- Tech Used: Python, Django, JavaScript, Tailwind CSS, SQLite, ThingSpeak.

IoT-Based Flood Monitoring Prototype

- Developed a field-tested IoT flood monitoring prototype using NodeMCU ESP8266 with a 16-meter sonar sensing range and GSM updates every 5 seconds.
- Integrated solar power for off-grid deployment and visualized telemetry via ThingSpeak cloud dashboards.
- Tech Used: Embedded C, NodeMCU ESP8266, Sonar Sensor, GSM Module, Solar Panel, ThingSpeak.

Education

Girijananda Chowdhury University, Btech in Electronics and Communication Engineering

Sept 2022 – May 2026

Kendriya Vidyalaya Lumding, Completed my Higher Studies

May 2019 – March 2021